

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

1192521055)

I Sugay Krishnan.P (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Descriptive Written Test

1.

Smart Home automation System.

Introduction:

It is used in sensors and AI to control lighting, temperatures and security based on environmental conditions and preference.

Types of Intelligent Agent:

There are 5 types of Agents.

1. Simple reflex Agent:

→ Act based on current input using predefined values.

→ Advantages: Fast and Simple

Eg: ~~at the~~

Switch on the light when the motion is detected.

2. Model based Agent:

→ Maintains an internal state of environment.

→ Advantages: It makes informed decision using previous observations.

Eg: Room occupancy and

3. Goal Based Agent.

- Selects actions to achieve goals
- Advantages: Support planning and decision making.

Eg: Maintains comfortable temperature and secures doors.

4. Utility Based agent:

- Select actions that maximize overall benefit.
- Advantages: To conflict the movement of room temperature.

Eg: Balance conflict security and electricity consumption.

2.

Uniform cost Search (UCS):

- Insert starting position into a priority queue
- Check whether it is the exit.
- Expand neighbouring position and update costs.
- Select lowest cost position.
- Repeat until exit found.

Definition: Deepening Search (IDS)

- Start with Depth Limit zero.
- Search up to the current limit
- Increase limit if ~~limit~~ Not found.
- Repeat until the exit reached.
- Advantages: Uses less memory and it is optimal when all step costs are equal.

Complexity Comparison

- Algorithms are UCS and IDS
- Time complexity: Exponential in worst case and $O(b^d)$.

Advantages: Compares alternative choices to the best action.

Conclusion:

The hybrid approach that improves the energy efficiency, security personalization by combining decision-making.

Robot Navigation in Unknown maze using UCS and IDS:

Introduction:

A Robot can use uniformed search in algorithms to explore an unknown maze and find path to exit.

Types of Agents:

- Simple Reflex
- Model-Based
- Goal-Based
- Utility-Based

Conclusion:

UCS minimized the paths cost, while IDS reduces usage combining a model-based remember discovered passage.